

Reminders

Upcoming due dates

Wed Mar 04 EDA checkpoint

Fri Mar 06 D8

How to be wrong

Data Science in Practice

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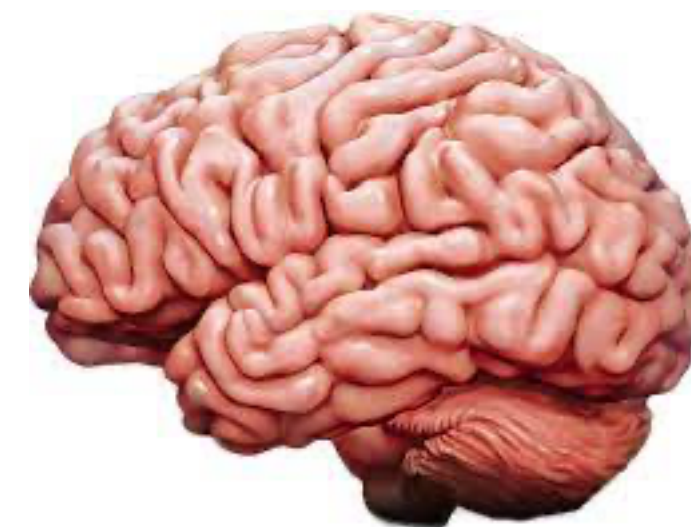
Errors of measurement

Errors of analysis

Errors of broken tools

Errors of human cognition

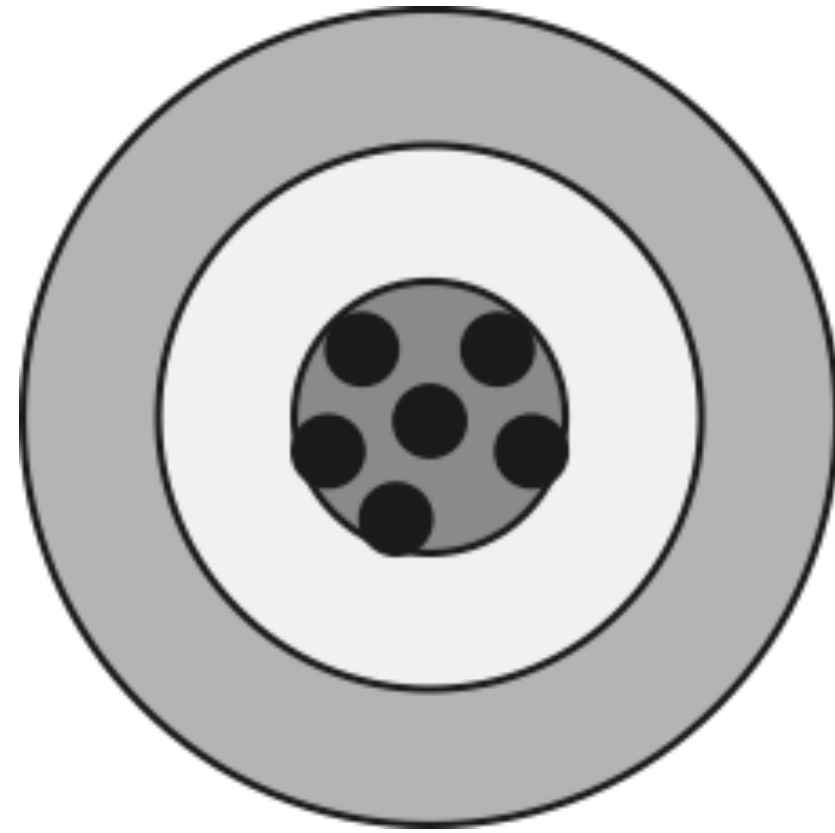
Errors of communication



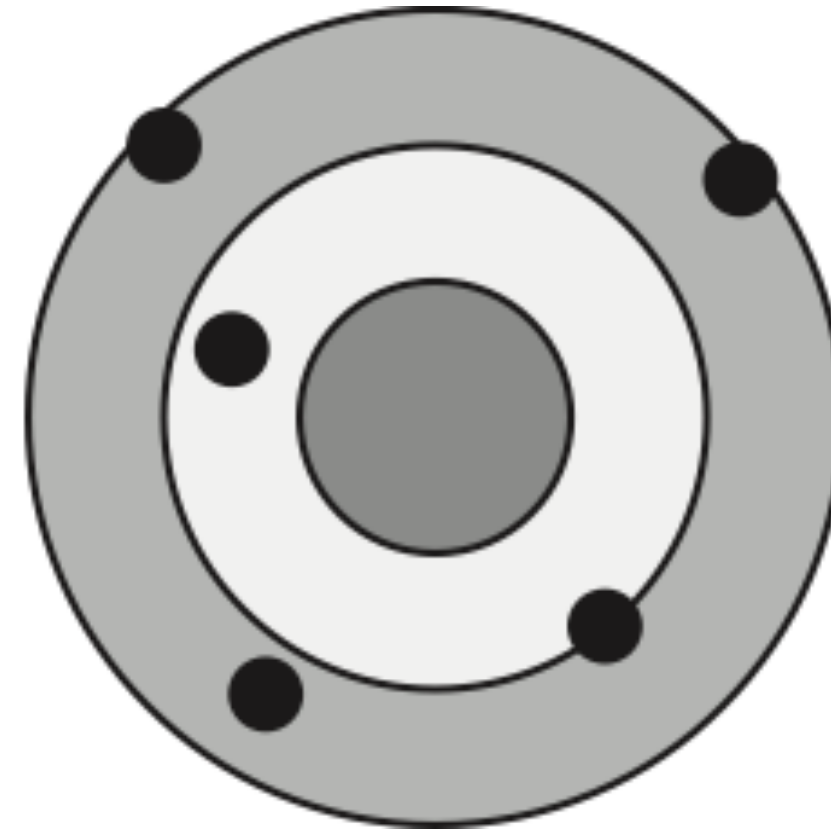


Errors of measurement

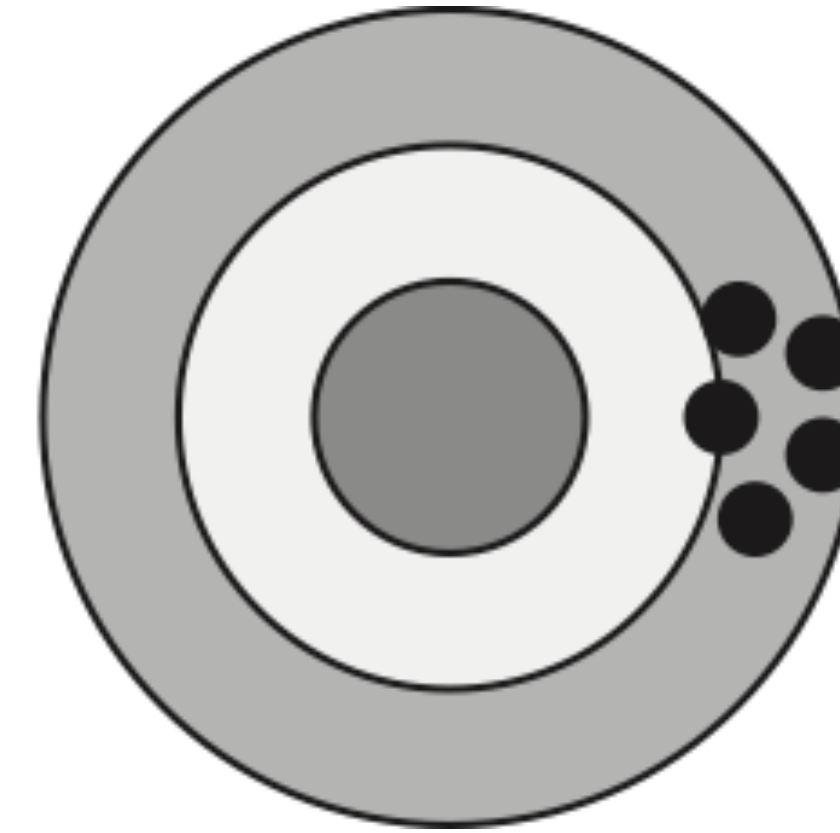
Accurate
and precise



Accurate
but not precise

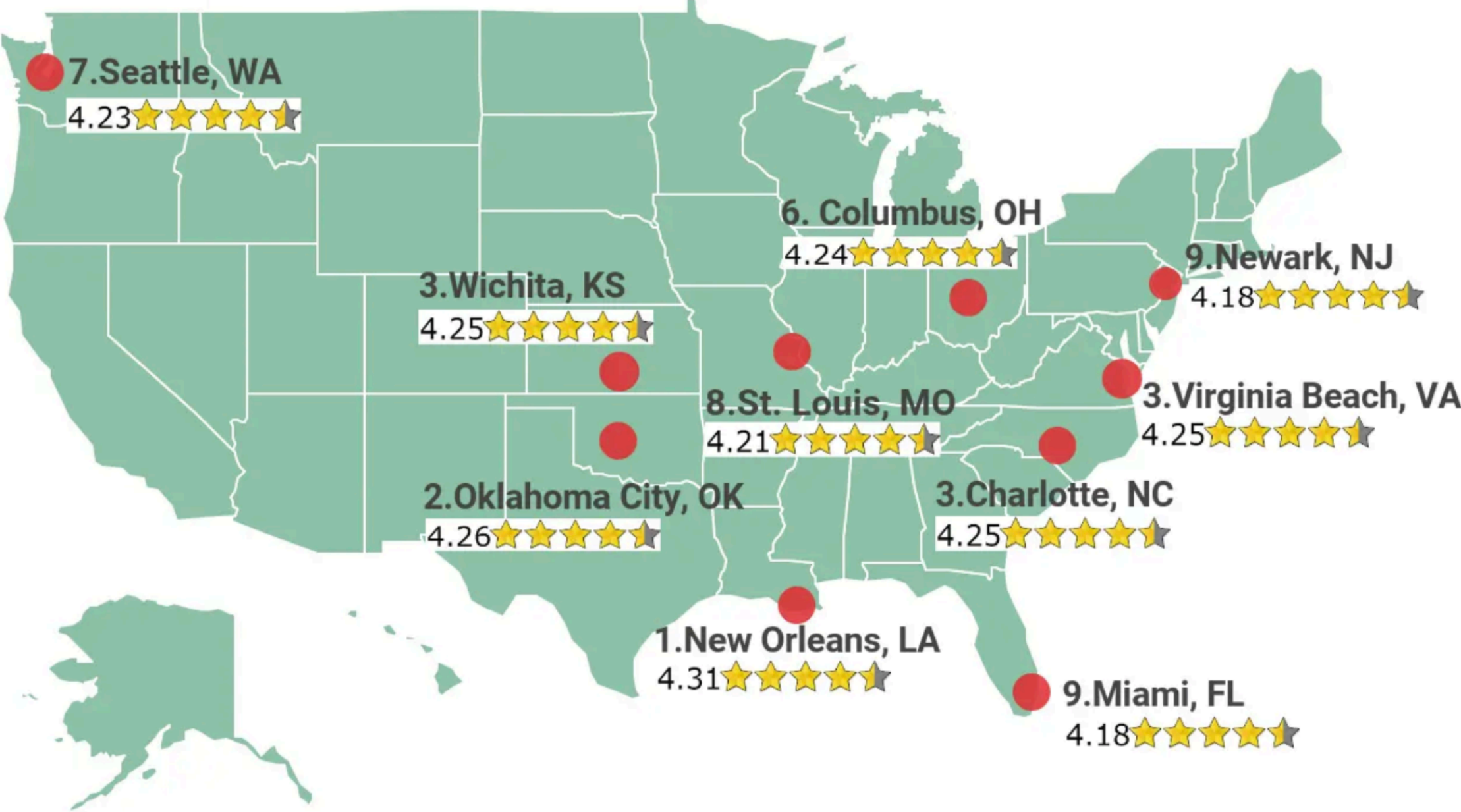


Precise
but not accurate



Top Cities For BBQ in the U.S.

An analysis of TripAdvisor restaurant reviews by chefspencil.com



Worst Cities For BBQ in the U.S.

An analysis of TripAdvisor restaurant reviews by chefspencil.com



Proxy measurements

We can't measure directly what we care about, but we can measure something related

- Unemployment rate -> our general economy
- Gross domestic product -> standard of living
- BMI -> health
- Your suggestions?
- Grades -> learning

**Do we understand what we are
measuring?**

Abraham Wald

Survivorship bias

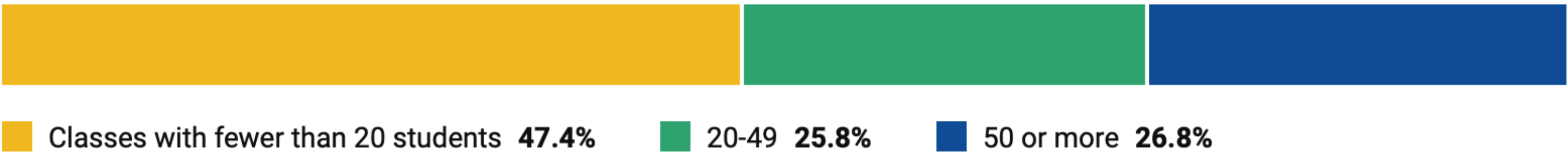


Are we measuring what's relevant?

Academic Life at University of California--San Diego

The student-faculty ratio at University of California--San Diego is 19:1, and the school has 47.4% of its classes with fewer than 20 students. The most popular majors at University of California--San Diego include: Biology, General; Mathematics; Economics; International/Global Studies; and Computer Science. The average freshman retention rate, an indicator of student satisfaction, is 94%.

Class Sizes

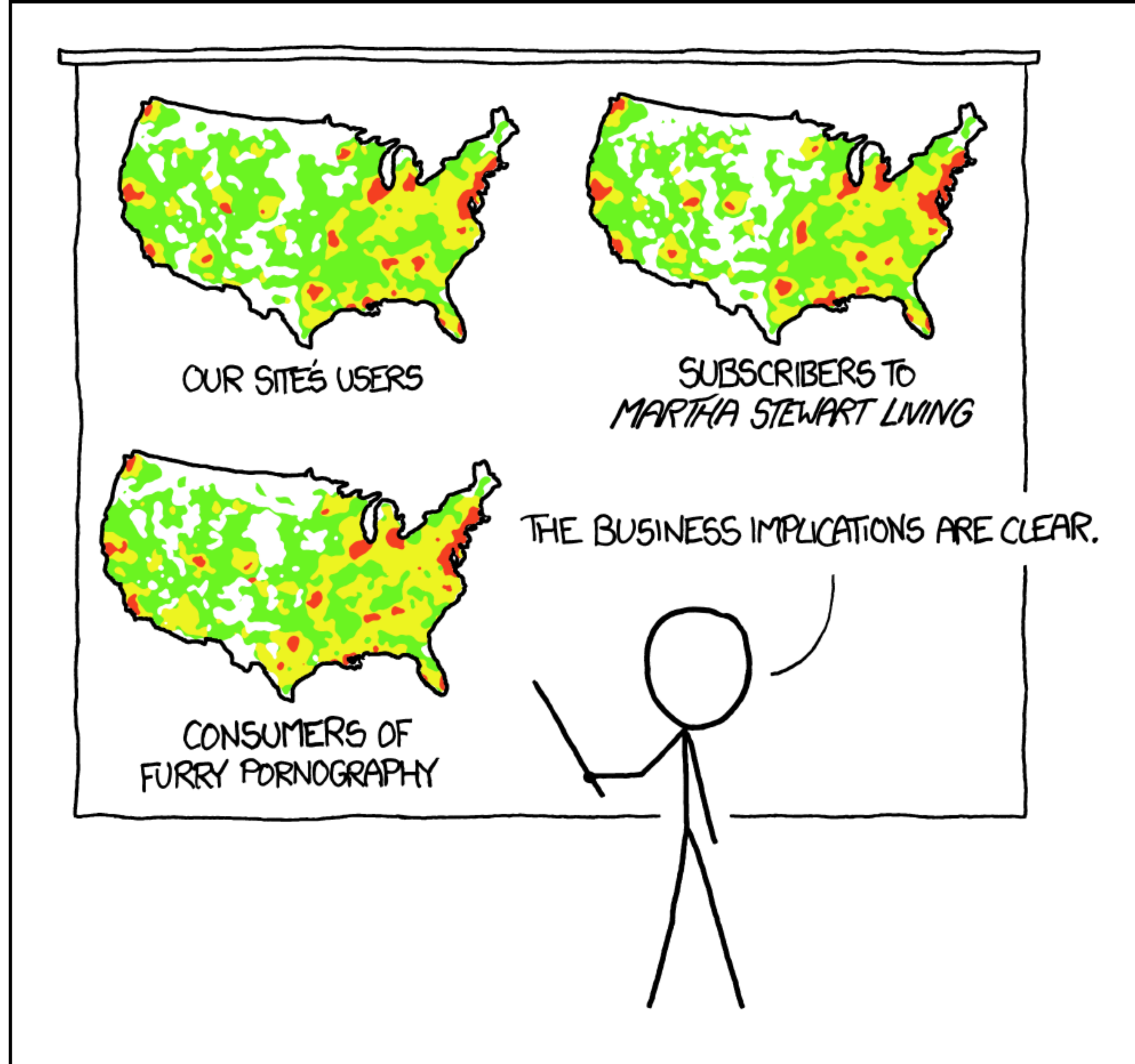


Student-faculty ratio	19:1
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4-year graduation rate	65%
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UCSD median class size vs median student experience

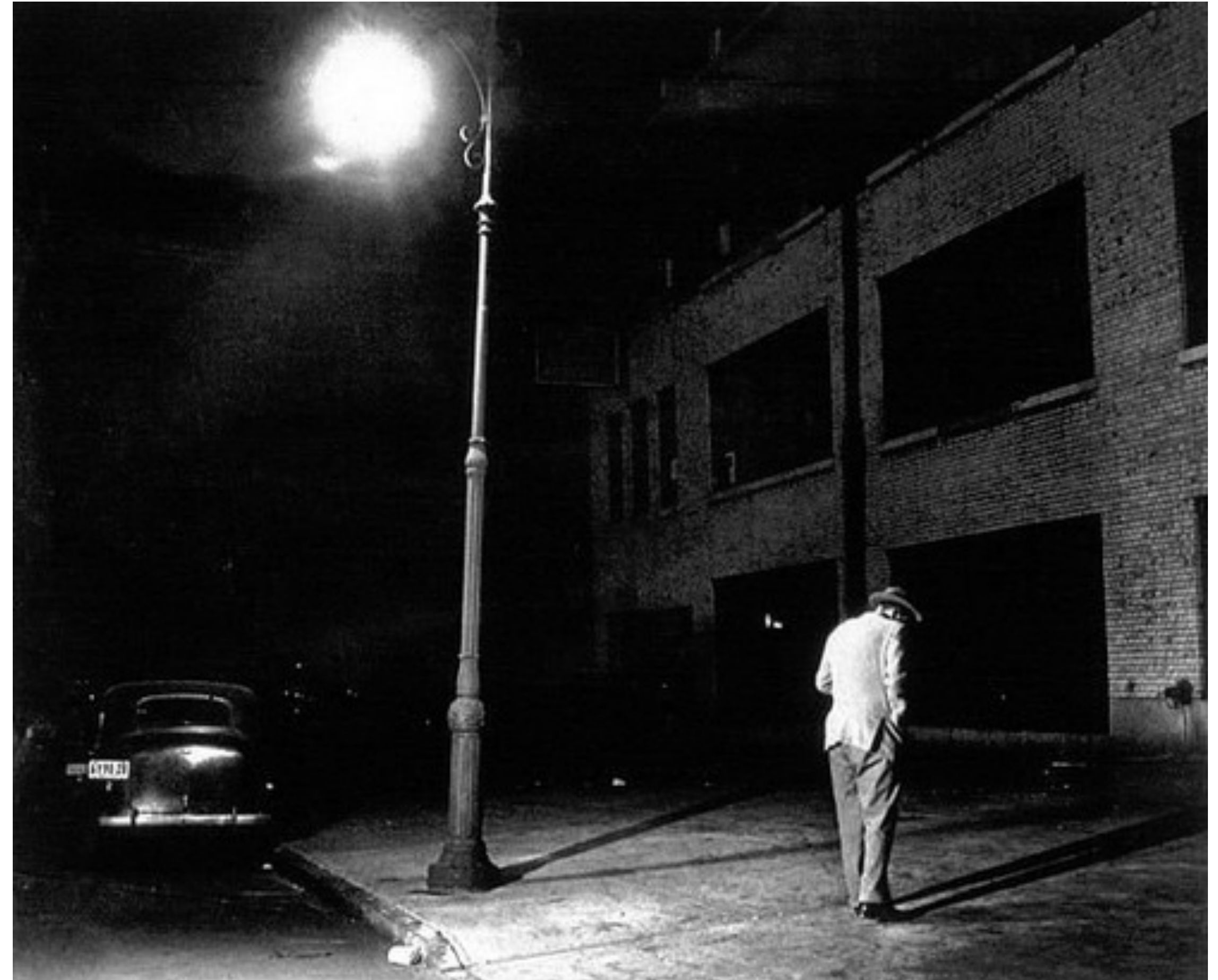
		% of classes with this many students	Cumulative %	Fraction of classes with this many students * min number of students in that class type	% of students in these classes (normalized version of column to the left)	
	2-9 students:	12%	12%	0.24	0.67%	
	10-19 students	32%	44%	3.2	8.95%	
Median class size as experienced by faculty	20-29 students:	14%	58%	2.8	7.83%	
	30-39 students:	8%	66%	2.4	6.72%	
	40-49 students:	4%	70%	1.6	4.48%	
	50-99 students:	11%	81%	5.5	15.39%	
Median class size as experienced by students	Over 100 students:	20%	101%	20	55.96%	
			Sum:	35.74		
	Data from https://www.collegedata.com/college/University-of-California-San-Diego/					



PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE
BASICALLY JUST POPULATION MAPS

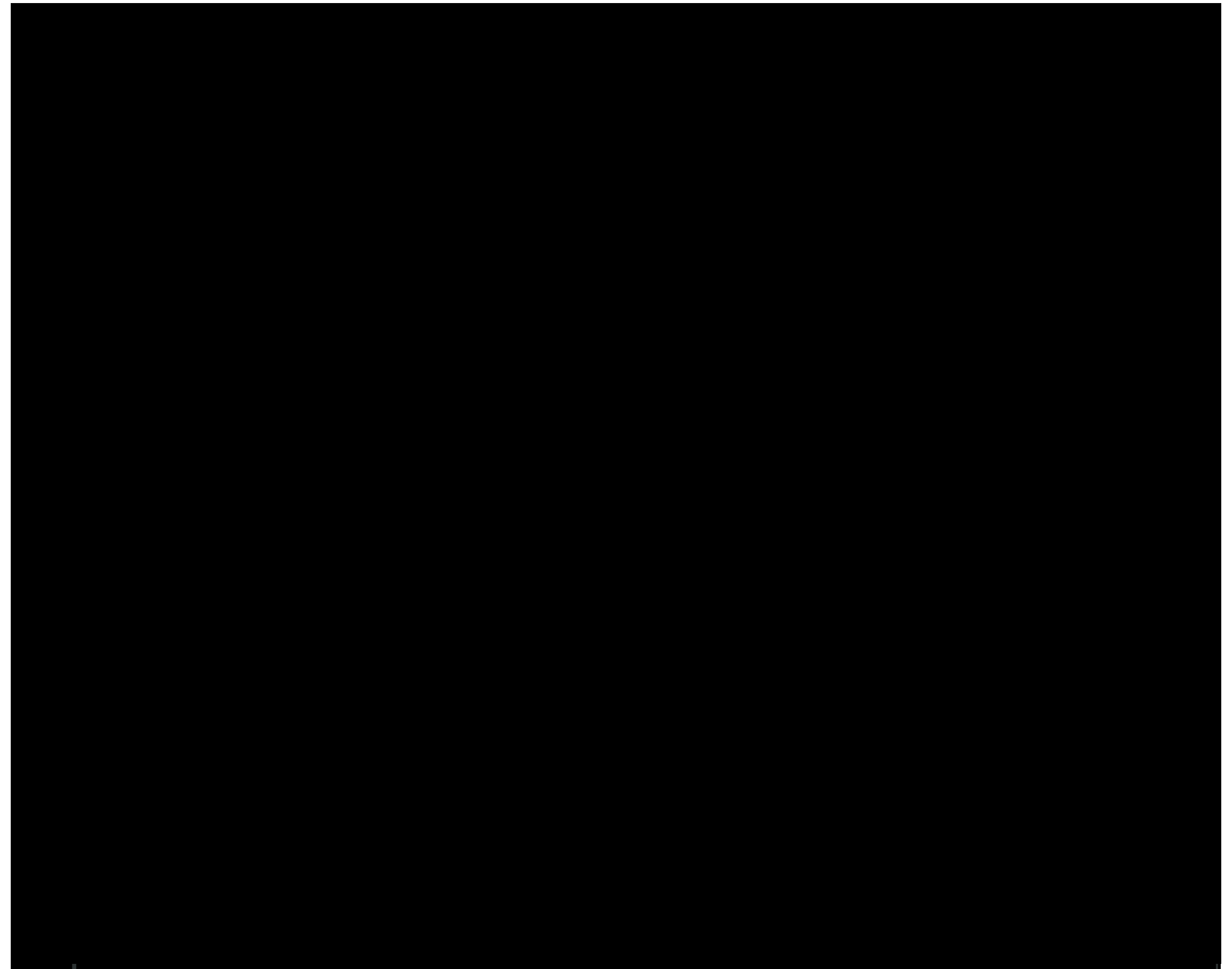
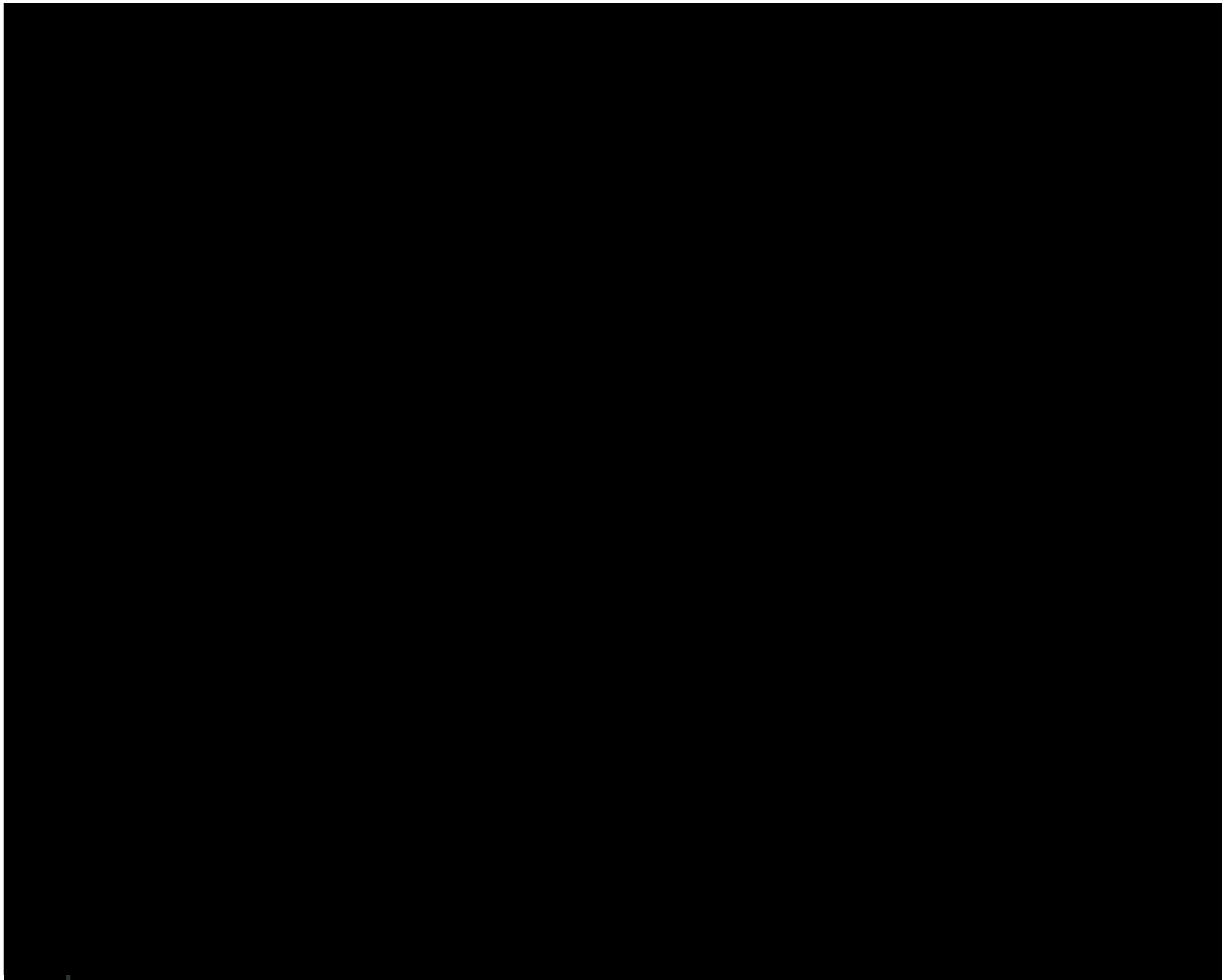
The Lamppost Problem

aka. Streetlight Effect
aka. The Drunkard's Search
aka. Picking a shitty proxy





Many, and possibly most, scientists spend their careers looking for answers where the light is better rather than where the truth is more likely to lie.



Goodhart’s law

Once a metric becomes a goal, you’re fucked

Volkswagen emissions scandal

Article

Talk

Read

Edit

View history

Tools

From Wikipedia, the free encyclopedia

"Dieselgate" and "Emissionsgate" redirect here. For other diesel emissions scandals, see Diesel emissions scandal.

The **Volkswagen emissions scandal**, sometimes known as **Dieselgate** or **Emissionsgate**, began in September 2015, when the **United States Environmental Protection Agency** (EPA) issued a notice of violation of the **Clean Air Act** to German automaker **Volkswagen Group**. The agency had found that Volkswagen had intentionally programmed **turbocharged direct injection** (TDI) **diesel engines** to activate their **emissions** controls only during laboratory **emissions testing**, which caused the vehicles' **NO_x** output to meet US standards during regulatory testing. However, the vehicles emitted up to 40 times more NO_x in real-world driving. Volkswagen deployed this software in about 11 million cars worldwide, including 500,000 in the United States, in **model years** 2009 through 2015.

Volkswagen emissions scandal



A 2010 Volkswagen Golf TDI displaying "Clean Diesel" at the **Detroit Auto Show**

Search engine optimization

Article

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Tools

From Wikipedia, the free encyclopedia

"SEO" redirects here. For other uses, see Seo.

Search engine optimization (**SEO**) is the process of improving the quality and quantity of **website traffic** to a **website** or a **web page** from **search engines**. SEO targets unpaid traffic (known as "natural" or "**organic**" results) rather than direct traffic or **paid traffic**. Unpaid traffic may originate from different kinds of searches, including **image search**, **video search**, **academic search**, news search, and industry-specific **vertical search** engines.

As an **Internet marketing** strategy, SEO considers how search engines work, the computer-programmed **algorithms** that dictate search engine behavior, what people search for, the actual search terms or **keywords** typed into search engines, and which search engines are preferred by their targeted audience. SEO is performed because a website will receive more visitors from a search engine when websites rank higher on the **search engine results page** (SERP). These visitors can then potentially be converted into customers.

- Part of a series on
- Internet marketing**
- Search engine optimization**
- Local search engine optimisation
- Social media marketing
- Email marketing
- Referral marketing
- Content marketing
- Native advertising

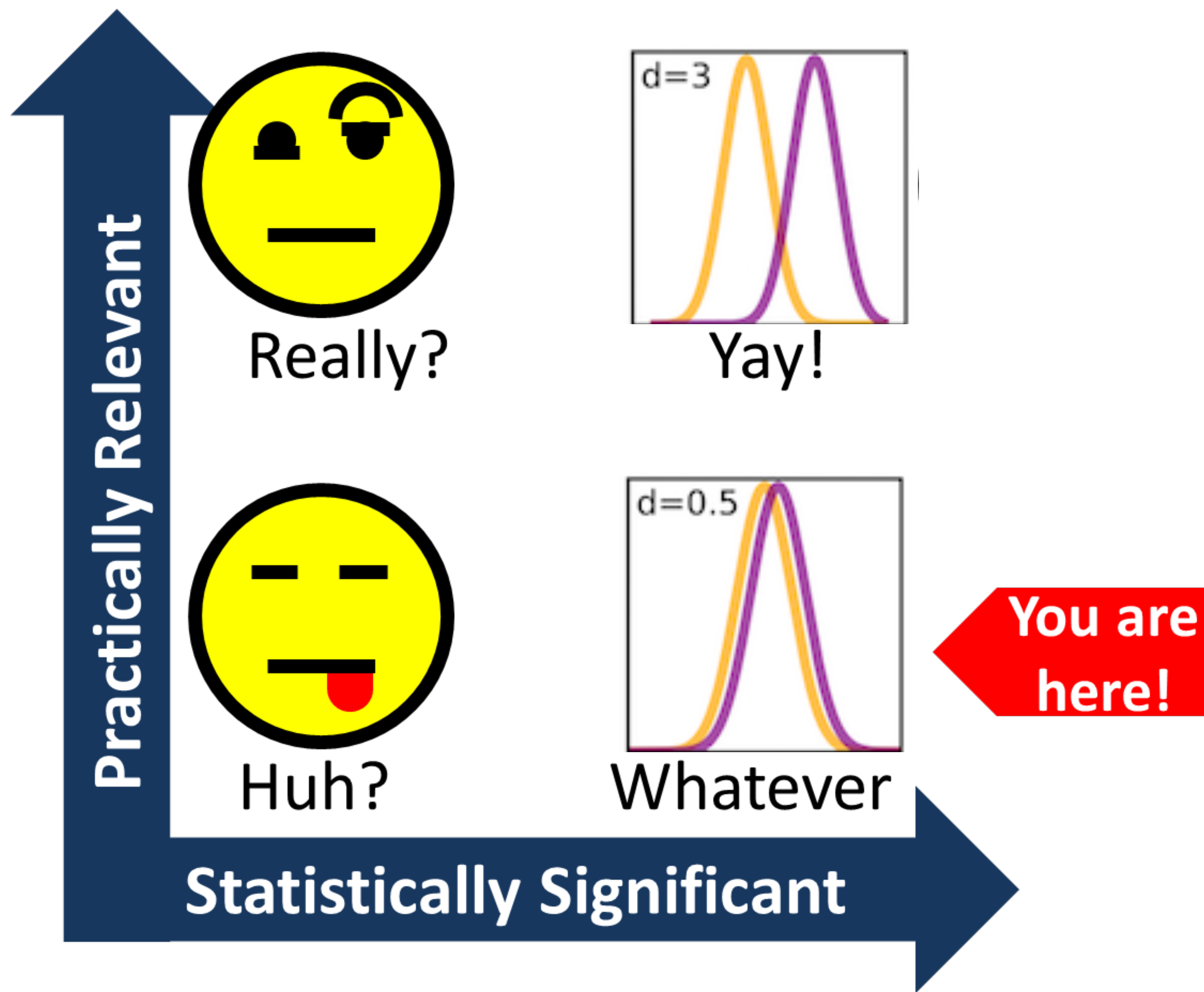
Search engine marketing

- Pay-per-click
- Cost per impression
- Search analytics
- Web analytics

Display advertising

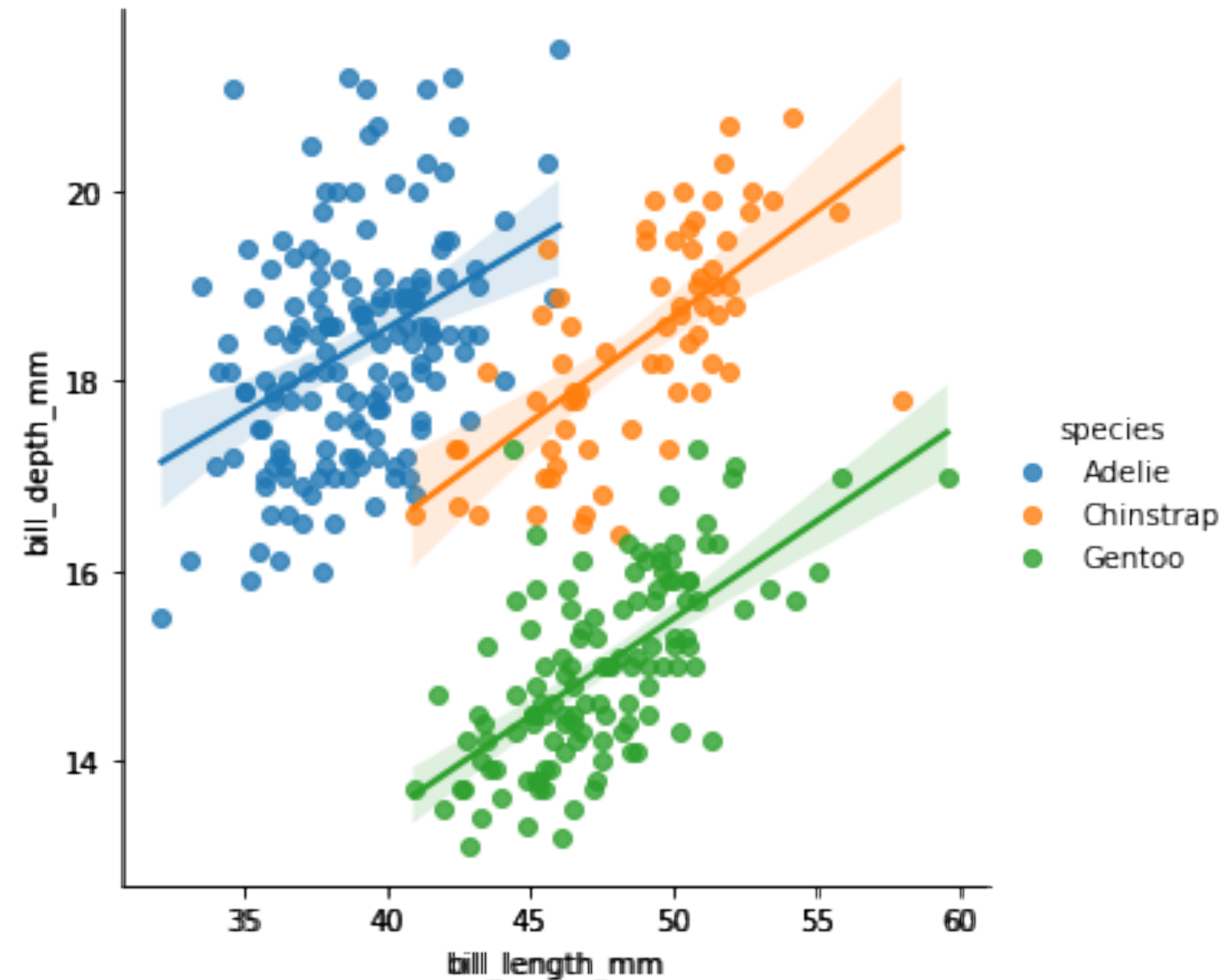
- Ad blocking
- Contextual advertising

Errors of analysis

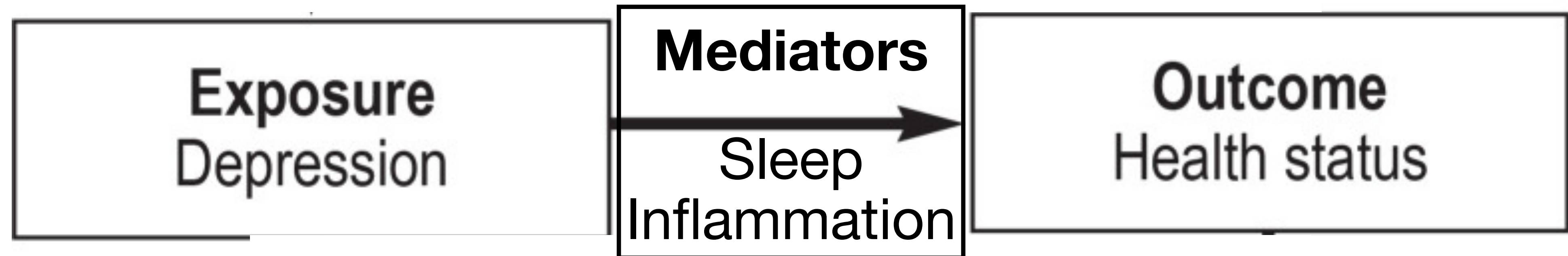


Simpson's paradox

Answers change when you include subgroups in your analysis



Confounders, Colliders, and Mediators



Confounders and Colliders and Mediators

- Confounders: $X \leftarrow Z \rightarrow Y$ aka you made the wrong model the first time
- Colliders: $X \rightarrow Z \leftarrow Y$ aka selection bias
- Mediators: $X \rightarrow Z \rightarrow Y$ aka science is hard and advances in small steps
- To estimate the total effect of exposure X on outcome Y :
 - DO adjust for confounders (common causes of X and Y)
 - DO NOT adjust for colliders (common effects of X and Y) but you should be aware of them because you want to avoid them in your study design
 - DO NOT adjust for mediators (variables on the causal pathway)

Adjusting for a Confound

```
import statsmodels.api as sm
import statsmodels.formula.api as smf
import pandas as pd
import numpy as np

# 1. Create a sample dataset
# Confounder: Age (affects both Smoking and LungCapacity)
# Exposure: Smoking (0=No, 1=Yes)
# Outcome: LungCapacity
np.random.seed(42)
n = 100
age = np.random.randint(20, 80, n)
smoking = (age > 50).astype(int) + np.random.randint(0, 2, n) # Higher age = more smoking
smoking = np.clip(smoking, 0, 1)
lung_capacity = 100 - (0.5 * age) - (10 * smoking) + np.random.normal(0, 5, n)

df = pd.DataFrame({'age': age, 'smoking': smoking, 'lung_capacity': lung_capacity})

# 2. Unadjusted Model (Simple Regression)
# Shows the raw relationship between smoking and lung capacity
model_unadjusted = smf.ols('lung_capacity ~ smoking', data=df).fit()
print("Unadjusted Effect of Smoking:\n", model_unadjusted.params['smoking'])

# 3. Adjusted Model (Multiple Regression)
# Includes 'age' to control for its confounding effect
model_adjusted = smf.ols('lung_capacity ~ smoking + age', data=df).fit()
print("\nAdjusted Effect of Smoking (holding age constant):\n", model_adjusted.params['smoking'])
print("\nFull Model Summary:\n", model_adjusted.summary())
```

... Unadjusted Effect of Smoking:
-20.25200142142117

Adjusted Effect of Smoking (holding age constant):
-10.442326161410248

Full Model Summary:

OLS Regression Results						
Dep. Variable:	lung_capacity		R-squared:	0.854		
Model:	OLS		Adj. R-squared:	0.851		
Method:	Least Squares		F-statistic:	283.6		
Date:	Mon, 02 Mar 2026		Prob (F-statistic):	3.02e-41		
Time:	17:43:29		Log-Likelihood:	-300.09		
No. Observations:	100		AIC:	606.2		
Df Residuals:	97		BIC:	614.0		
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	98.4348	1.452	67.795	0.000	95.553	101.317
smoking	-10.4423	1.265	-8.257	0.000	-12.952	-7.932
age	-0.4665	0.033	-14.188	0.000	-0.532	-0.401
Omnibus:	2.366	Durbin-Watson:	2.021			
Prob(Omnibus):	0.306	Jarque-Bera (JB):	1.876			
Skew:	-0.324	Prob(JB):	0.391			
Kurtosis:	3.174	Cond. No.	155.			

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

<https://forms.gle/sPwQg3szQKY8tegr8>

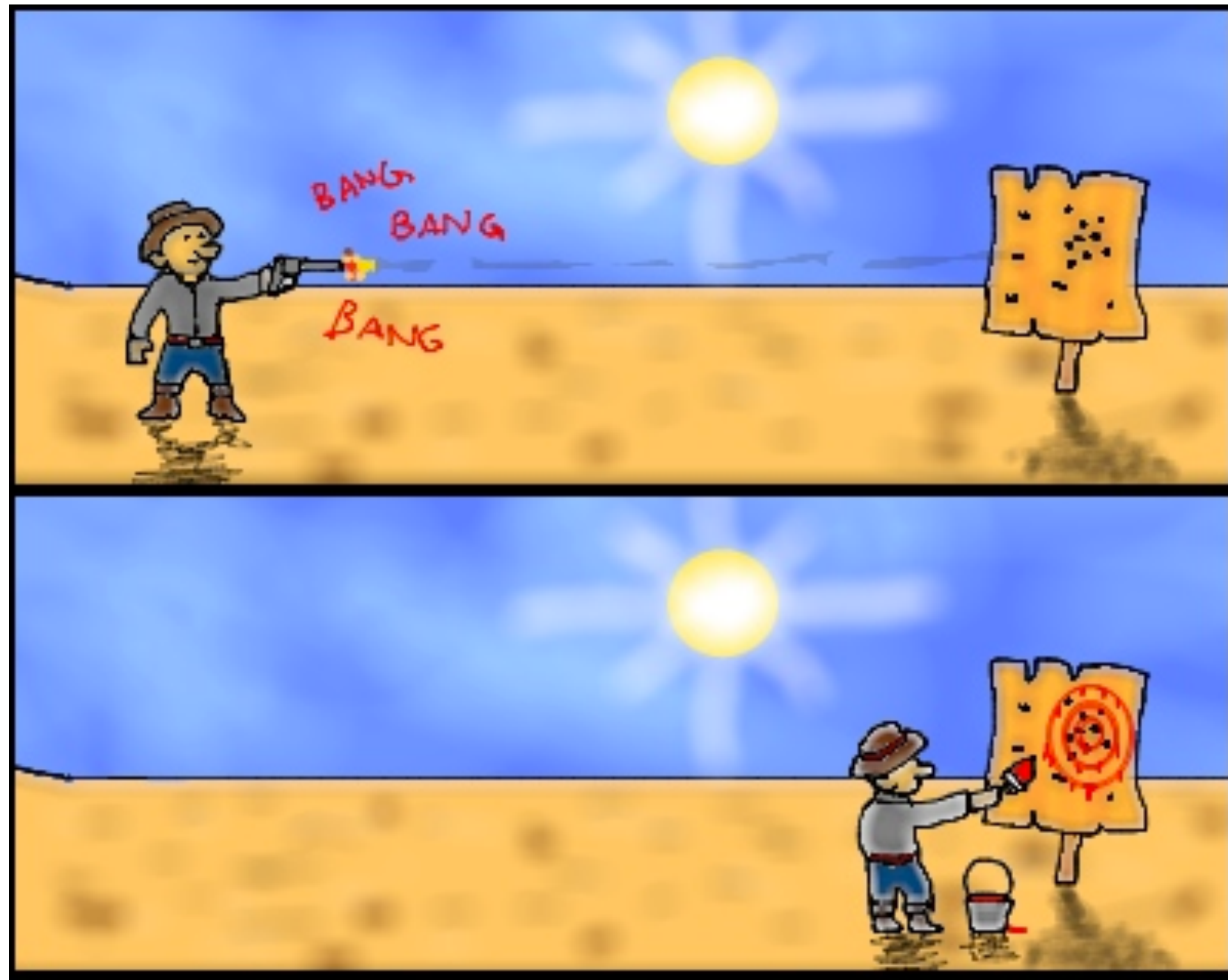


Bad methods



The Texas Sharpshooter fallacy is characterized by a lack of a specific hypothesis prior to the gathering of data, or the formulation of a hypothesis only after data have already been gathered and examined.

Avoiding HARKing



- Best possible:
 - EDA on a separate dataset than you do your NHST stuff
 - Build up knowledge in multiple studies that do the same thing
- In many DS contexts this is impossible. Just be clear that p-values once again are suggestive not definitive